MIL-P-46489C (PA)
4 JANUARY 1974
SUPERSEDING
MIL-P-46489B (MU)
25 April 1969

# MILITARY SPECIFICATION

## PROPELLANT M30
## FOR USE IN CARTRIDGE, 105MM, APDS-T, M392A2

### 1. SCOPE

1.1 This specification covers M30 Propellant for use in Cartridge, 105MM, APDS-T, M392A2.

### 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-A-2550- | Ammunition and Special Weapons; General Specification for |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-109 - | Quality Assurance Terms and Definitions |
| MIL-STD-652 - | Propellants Solid for Cannons Requirements and Packing |
| MIL-STD-1168- | Lot Numbering of Ammunition |
| MIL-STD-1171- | Acceptance and Description Sheets for Propellants and Explosives |
| MIL-STD-414 - | Sampling Procedures and Tables for Inspection by Variables for Percent Defective |

DRAWINGS

ARMY

FSC: 1375

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8863427 - Cartridge, 105MM, APDS-T, M392A2, Assembly and Marking Diagram

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

Title 49, Department of Transportation: Parts 0-199

AMC PAMPHLET 706-114

Experimental Statistics, Section 5, Tables

The Code of Federal Regulations is available from the Superintendent of Documents, Government Printing Office, Washington, D.C., 20402. Orders for the above publication should cite: 49 CFR 0-199 (latest revision).

3. REQUIREMENTS

3.1 Materials. The propellant shall be M30, Type I conforming to the requirements of MIL-STD-652.

3.2 Proving Ground. The propellant, when fired in the M68 gun and with the weight of projectile (12.94 pounds (lbs) ) for which it was manufactured, shall comply with the following requirements when tested as specified in 4.3.

3.2.1 Assessment. At 70 degrees Fahrenheit (°F) and standard weapon conditions, the propellant shall be capable of charge assessment to a velocity of 4875 feet per second (fps) without an associated expected pressure greater than 61,800 pounds per square inch (psi).

3.2.2 Uniformity of velocity. The standard deviation in velocity for the lot at 70°F shall not exceed 12.0 fps.

3.2.3 Permissible Individual Maximum Pressure. The propellant at recommended charge weight shall not produce a chamber pressure greater than 75,700 psi at any operating temperature (-50°F to +145°F) when fired in a standard weapon.



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3.3 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. No reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification unless otherwise specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inpsection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109, in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product. At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisons specified within this specification.

b. Number of batches of product inspected (shall be made available upon request by the contracting officer).

c. Results obtained, by defect code, for all inspections performed (shall be made available upon request by the contracting officer)

d. Drawing, specification number, and date,together with an identification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in



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any of the contracting documents (shall be made available upon request by the contracting officer).

f. Number of items in the lot.

g. Date submitted.

h. Eight copies of propellant description sheet SMU Form 1047 containing the applicable data (see 6.8)

4.1.2 The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation. The term "lot" as used throughout this specification, refers to any inspection lot, which is defined as an essentially homogeneous quantity of propellant from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. A lot shall consist of one or more sub-lots or one or more batches of M30 propellant produced by one manufacturer, in accordance with the same specification and same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of M30 propellant which has been subjected to the same unit chemical or physical process intended to make the final product homogeneous.

4.2.2 Testing

4.2.2.1 Materials (see 3.1). Each lot of propellant shall be sampled and tested for composition, form, dimensions and chemical-physical properties in accordance with MIL-STD-652.

4.2.2.2 Proving ground (see 3.2) Each lot of propellant shall be sampled for proving ground tests in accordance with MIL-STD-652. All tests shall be conducted using M68 tubes having a remaining life of at least 90 percent (see 6.12)

4.2.2.2.1 Assessment (see 3.2.1) Major defect, Code No. 01001 (see 6.4) The lot shall be rejected if the assessed charge weight of propellant has an expected pressure greater



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than 61,800 psi.

4.2.2.2.2 Velocity Uniformity. (70°F) (See 3.2.2) Major defect, Code No. 02001. If the product of the factor (see 6.5) times the velocity standard deviation of the rounds tested in accordance with 4.3.3 exceeds the applicable requirement the lot shall be rejected. For initial production lots a ten round sample shall be tested. (see 6.6).

4.2.2.2.3 Individual pressure (see 3.2.3)

4.2.2.2.3.1 Initial production. If any individual pressure observed during uniformity firings at (-50°F to 145°F) after correction for difference between as fired and recommended service charge weight exceeds 75,700 psi, the lot shall be rejected.

4.2.2.2.3.2 Subsequent Production 70°F. If any individual pressures observed during the 70°F uniformity series when corrected to standard conditions and recommended charge weight exceeds 75,700 minus 75 times the pressure temperature coefficient established during Initial Production Testing, the lot shall be tested at +145°F in accordance with 4.3.3.1.1.

4.3 Test methods and procedures

4.3.1 Proving ground tests. These tests shall be conducted at a Government proving ground (designated by the contracting officer) in accordance with the applicable Proving Ground Acceptance Test Procedure.

4.3.2 Charge assessment

4.3.2.1 Charge Slope Method. Three or more groups of rounds loaded with selected charge weights of propellant (weights selected should bracket expected service charge weight) shall be conditioned and fired at 70°F. From the data obtained, a charge weight shall be calculated which will yield the prescribed service velocity. This calculated charge weight shall then be loaded into rounds for the uniformity series (see 4.3.3). The difference between the corrected velocity obtained during the uniformity series and the service velocity shall be compensated for by a final adjustment in charge weight using the slope of the velocity charge weight curve previously established.

4.3.2.2 Constant Slope Method. After ten(10) lots of a given production series have been fired to provide



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a good knowledge of the propellant characteristics, the charge slope method of assessment may be discontinued and the charge weight determined by the constant slope method. A composite velocity versus charge weight versus pressure curve can be established from test results generated. This curve can then be used as in 4.3.2.1 above to calculate the charge weight needed to achieve service velocity. This calculated charge weight is then to be loaded into rounds for the uniformity series and final adjustment is to be made as indicated above for the charge slope mehtod.

4.3.3 Uniformity series

4.3.3.1 Initial production

4.3.3.1.1 145°F. Ten rounds of M392A2 Cartridges shall be assembled with the assessed charge of propellant, conditioned to 145°F (as indicated by a thermocoupled control round (rnd) and fired alternately with calibration rounds. The velocity and pressure means as well as the standard deviations associated with these parameters shall be calculated. All results shall be corrected to recommended charge weight conditions (i.e. difference between as fired charge weight and recommended charge weight). (see 6.10)

4.3.3.1.2 70°F. Ten rounds of M392A2 Cartridges shall be assembled with the assessed charge of propellant, conditioned to 70°F (as indicated by thermocoupled control rnd) and fired alternately with calibration rounds. The velocity and pressure means, as well as the standard deviations associated with these parameters shall be calculated. All results shall be corrected to standard shell weight (12.94 lbs) and standard weapon/firing conditions (i.e. the difference between calibration performance under test conditions and standard conditions) and to recommended charge weight conditions.

4.3.3.1.3 -50°F. Ten rounds of M392A2 Cartridges shall be assembled with the assessed charge of propellant, conditioned to -50°F (as indicated by a thermcoupled control rnd) and fired alternately with calibration rounds. The velocity and pressure means as well as the standard deviations associated with these parameters shall be calculated. All results shall be corrected to recommended



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charge weight conditions (i.e. difference between as fired charge weight and recommended charge weight).

4.3.3.2 Subsequent Production. Seven rounds of M392A2 Cartridges shall be assembled and tested at 70°F while 145°F and -50°F testing shall be omitted entirely. The velocity and pressure means as well as the standard deviations associated with these parameters shall be calculated. All results at 70°F shall be corrected to standard shell weight (12.94 lbs) and standard weapon/firing conditions and recommended charge weight conditions (see 4.2.2.2.3.2)

4.3.4 Refiring. If for any reason the test agency or commodity center considers that test conditions have detrimentally affected the test results, addtional charges as required shall be tested.

## 5. PREPARATION FOR DELIVERY

5.1 Packing and marking. Packing and marking shall be in accordance with MIL-STD-652.

## 6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Packing required (see 5.1)
c. Process. Details of the manufacturing process and the equipment used by the Contractor will be submitted to the procuring activity in writing prior to commencement of manufacture.

6.2 Standard deviation. The standard deviation shall be calculated with (n-1) as the devisor in a standard statistical technique equivalent to that shown in MIL-STD-414, Section B, Example B-1.

6.3 Submission of test data. In addition to the normal distribution of records, when the propellant is procured by the Department of the Army, one copy of all ballistic test data shall be forwarded to: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-P, and one copy to ATTN: SARPA-ADEP-2, Dover, New Jersey 07801.

6.4 Inspection code number. The five digit code numbers assigned to the inspections herein are to faciliate future data collection and analysis by the Government.



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6.5 Factors. The factors, as taken from Table A-21 of AMC Pamphlet 706-114 using the 95 percent upper confidence limit (A.05) and n-1 degrees of freedom, make allowances for the probability that a standard deviation from a smaller sample size exceeds the true standard deviation by chance alone.

6.6 Initial production lots. The initial production lots represent the first three consecutive lots from each manufacture which have been produced and met the acceptance criteria of this specification. Lapse in production of 90 days or more shall cause testing to revert back to intial production plan.

6.7 Tolerance. A tolerance of -5°F applies to all -50°F and 145°F requirements and a tolerance of plus or minus 2°F to all 70°F requirements.

6.8 Submission of SMU Form 1047. Information contained in propellant description sheet, SMU Form 1047, need not be resubmitted in the form of a certificate provided the provisions regarding certifying of responsible agents is adhered to (see 4.1.2).

6.9 Surveillance sample. Samples shall be forwarded to Picatinny Arsenal, Dover, N.J. ATTN: SARPA-FR-G, for the propellant surveillance test program in accordance with MIL-STD-652.

6.10 Test cartridges shall be assembled in accordance with drawing 8863427.

6.11 Proving ground test summary

INITIAL PRODUCTION

| Temp (°F) | Sample Size | Vel(fps) (assessed) | Lot Std Dev of Vel (fps) | Perm Ind Max Pres (psi) | Avg. Pres Max (psi) |
|---|---|---|---|---|---|
| -50 | 10 | NA | NA | 75,700 | NA |
| 70 | 10 | 4875 | 12.0 | 75,700 | 61,800 |
| 145* | 10 | NA | NA | 75,700 | NA |



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SUBSEQUENT PRODUCTION

| | | | | | |
|---|---|---|---|---|---|
| 70 | 7 | 4875 | 12.0 | 75,700 | 61,800 |
| 145* | 10 | NA | NA | 75,700 | NA |

*To be fired only when 70°F results indicate that marginal pressure performance may exist (see 4.2.2.2.3.2).

6.12 Tube wear. If the restriction on tube life/history places an undue burden on the Proving Ground, a factor of 0.8 fps/0.001 inch of wear may be used for tube wear.

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OMB Approval
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INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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